I'm on a time crunch, and I need to study for my Deep Neural Networks Exam by reviewing the homework. Here is the first homework and solutions: https://static.us.edusercontent.com/files/aO8NnPYdzjOpEyEphrp1j0nZ

I want you to: In my experience, Claude is especially good at parsing context; it might be better than ChatGPT.

1. Identify the 5 most important problems (count subparts as individual problems) I meant modified (typo).
2. Ask me each question with its entire context, preferably barely unmodified, one at a time
3. I will respond with a conceptual approach (I don't have time to fully solve)
4. If I am correct, output the entire solution and move on to the next question; otherwise, continue guiding me to the correct approach

This is not something I could've asked of ChatGPT study mode, as it just tells me that it can't reveal solutions.

I'll fetch the homework and solutions document to help you study efficiently.

Fetched https://static.us.edusercontent.com/files/aO8NnPYdzjOpEyEphrp1j0nZ edusercontent.com

I provided the solutions directly to avoid hallucinated, incorrect solutions.

Perfect! I've analyzed the homework and identified the 5 most important problems based on conceptual depth and relevance to deep neural networks. Here they are:

The 5 Most Important Problems:

1. **Problem 1(e)** - Optimal learning rate for fastest convergence (introduces condition number concept) Whether these are the most "important" problems, I do not know, but I really like how it provided a short summary/justification for each.
2. **Problem 2(j)** - Proving SGD convergence with stochastic Lyapunov functions (core convergence theory)
3. **Problem 3(g)** - Momentum acceleration analysis (practical optimizer understanding)
4. **Problem 5(a)** - Instance noise as regularization (connects data augmentation to ridge regression)
5. **Problem 7** - Kernel Ridge form via MAP estimation (probabilistic interpretation)

Let me start with the first question:

Question 1: Problem 1, Part (e)

Context: You have a vector problem with two parameters w[1] and w[2], where one has a large σ_ℓ and the other has a tiny σ_s (i.e., σ_ℓ ≫ σ_s). You want to solve the equation:

One thing that Claude struggles with is making math look good. ChatGPT is way better at this, but this is still manageable.

$$\begin{bmatrix} \sigma_\ell & 0 \\ 0 & \sigma_s \end{bmatrix} \begin{bmatrix} w[1] \\ w[2] \end{bmatrix} = \begin{bmatrix} y[1] \\ y[2] \end{bmatrix}$$

using gradient descent with a single learning rate η starting from w = 0.

From the previous parts, you know:

- For convergence, you need $\eta < 1/\sigma^2_\ell$ (the large σ limits your learning rate)
- Each dimension converges at rate $|1 - 2\eta\sigma^2_i|$
- The overall convergence is dominated by the slower of the two dimensions

Question: The speed of convergence overall will be dominated by the slower of the two. For what value of η will we get the fastest overall convergence to the solution?

What is your conceptual approach to finding the optimal learning rate?

Since we want to maximize the minimum of the two, we can set the rates equal to each other and derive 2 solutions, one of them being $\eta = 0$.

Excellent! You've identified the key insight perfectly. You're absolutely right that to minimize the maximum convergence rate (which means fastest overall convergence), we want to balance the two dimensions so they converge at the same speed. This is a classic minimax optimization principle.

I like how it continued to explain, and didn't just move on.

Let me elaborate on why your approach is correct and what happens when we set the rates equal:

When you set $|1 - 2\eta\sigma^2_\ell| = |1 - 2\eta\sigma^2_s|$, you're looking for the point where both dimensions take equally long to converge. Since $\sigma_\ell \geq \sigma_s$, we know that $1 - 2\eta\sigma^2_\ell$ will become negative before $1 - 2\eta\sigma^2_s$ does as we increase η. This means we're really solving:

$-(1 - 2\eta\sigma^2_\ell) = (1 - 2\eta\sigma^2_s)$

Which gives us two solutions as you mentioned. The trivial solution $\eta = 0$ comes from when both sides equal 1, which doesn't help us make any progress. The non-trivial solution gives us the sweet spot where the large eigenvalue dimension (which wants a small learning rate) and the small eigenvalue dimension (which can tolerate a larger learning rate) are perfectly balanced.

This optimal learning rate turns out to be $\eta = 1/(\sigma^2_\ell + \sigma^2_s)$, which is actually the harmonic mean-like compromise between the two extreme values $1/\sigma^2_\ell$ and $1/\sigma^2_s$.

Here is the complete solution:

Solution: Recall that the minimum of the pointwise maximum of two functions occurs at a point where both functions are equal. Thus, the fastest convergence is achieved when $|1 - 2\eta\sigma^2_\ell| = |1 - 2\eta\sigma^2_s|$, and since $\sigma_\ell \geq \sigma_s$, there are two solutions: one is the trivial $\eta = 0$, and the other is:

$$\eta = \frac{1}{\sigma_\ell^2 + \sigma_s^2}$$

See Figure 1 for a picture showing the intersection of the two convergence rate curves. The optimal learning rate occurs at the non-trivial intersection where both dimensions converge at the same rate, achieving the fastest possible overall convergence.

Ready for the next question?

Yes.

The number of free messages you can send on Claude is pretty limited, so it was a bit annoying that I had to waste a message here, but I guess this let's me ask any remaining questions.

Question 2: Problem 2, Part (j)

Context: You're analyzing the convergence of Stochastic Gradient Descent (SGD) for an interpolating least-squares problem. Through a series of coordinate transformations, you've reduced the problem to analyzing a square system where you're trying to show that a stochastic Lyapunov function $L(\tilde{w}) = \tilde{w}^T\tilde{X}^T\tilde{X}\tilde{w}$ decreases exponentially in expectation.

You've already established two key results in the previous parts:

- Part (h) showed that the linear term A in the Lyapunov evolution satisfies: $E[A|\tilde{w}_t] \leq -c_1\eta L(\tilde{w}_t)$ where $c_1 = (4/n)\sigma^2\_min$
- Part (i) showed that the quadratic term B satisfies: $E[B|\tilde{w}_t] \leq c_2\eta^2 L(\tilde{w}_t)$ where $c_2 = (4/n)\sigma^2\_max\rho^2$

Claude did an incredibly good job summarizing the context from the previous parts of the question. It was as if I solved them already and could use them for this part.

You've shown that the Lyapunov function evolves as:
$L(\tilde{w}_{t+1}) = L(\tilde{w}_t) + A + B$

Putting these together gives you:
$E[L(\tilde{w}_{t+1})|\tilde{w}_t] \leq (1 - c_1\eta + c_2\eta^2)L(\tilde{w}_t)$

where both $c_1 > 0$ and $c_2 > 0$.

Question: Show that this means there exists a small enough η so that $1 - c_1\eta + c_2\eta^2 < 1$.

What is your conceptual approach to finding the condition on η that ensures contraction?

I have no idea what this is even asking.

I was honestly confused by the wording of this question, since I didn't get how the previous parts were relevant.

Let me break this down step by step - this is actually asking something simpler than it might appear!

What the question is really asking:

We have an inequality that looks like this: $(1 - c_1\eta + c_2\eta^2)$ times something. The question wants you to show that we can make the expression $(1 - c_1\eta + c_2\eta^2)$ be strictly less than 1 by choosing η

appropriately.

Think about what this expression represents. When $\eta = 0$ (no learning at all), we get $1 - 0 + 0 = 1$, meaning no progress. As we increase η slightly from zero, what happens?

The key insight:

Look at the three terms: we start at 1, then we subtract $c_1\eta$ (which is positive and grows linearly with η), then we add back $c_2\eta^2$ (which is positive but grows quadratically with η).

For small values of η, which term dominates - the linear term $c_1\eta$ or the quadratic term $c_2\eta^2$? The linear term dominates when η is small! This means that for small enough η, the subtraction of $c_1\eta$ "wins" over the addition of $c_2\eta^2$, making the whole expression less than 1.

Your task conceptually:

You need to show that there exists some range of η values where:

- The expression is less than 1 (so we get contraction and convergence)
- η is positive (so we actually make progress)

This is equivalent to finding when $1 - c_1\eta + c_2\eta^2 < 1$, which simplifies to: when is $-c_1\eta + c_2\eta^2 < 0$?

What condition on η makes $-c_1\eta + c_2\eta^2 < 0$?

Can you factor this and determine what range of η values would work?

This hint was great and clarified the problem without solving it.

This is basically just a quadratic equation then, and we can solve it to find the zeroes. But I'm confused because I don't get why we need the previous parts at all.

Here, the conversation pivoted to a very detailed explanation. Claude did super well breaking this down, especially because it summarized parts of the question before this.

Ah, excellent question! This reveals something really important about how mathematical proofs are structured, and I'm glad you're asking because it shows you're thinking critically about the logic.

You're absolutely right that solving the inequality is "just" a quadratic, but let me explain why the previous parts were essential - this gets at the heart of how convergence proofs work in optimization.

Why we needed the previous parts:

The previous parts weren't just busywork - they were constructing a *proof strategy*. Here's the big picture of what we accomplished:

We started with a complex stochastic process (SGD iterating on a system of equations) and we wanted to prove it converges. But how do you prove a random, iterative algorithm converges?

You can't just "solve" it directly. Instead, you use a **Lyapunov function** - think of it like an energy function that we hope decreases over time.

Parts (h) and (i) were crucial because they *bounded* how much this energy function changes at each step. Without those bounds, we wouldn't know that the constants c_1 and c_2 are well-defined, finite, and positive. We had to carefully analyze:

- Part (h): How much does the gradient step help us? (The A term - this is the "good" term that pulls us toward the solution)
- Part (i): How much does the randomness hurt us? (The B term - this is the "bad" term from stochastic noise)

The payoff in part (j):

Now that we've bounded both effects, part (j) asks you to verify that *for small enough learning rates*, the good effect (linear in η) dominates the bad effect (quadratic in η). This is the final step that proves SGD actually converges exponentially!

So yes, the algebra is simple quadratic manipulation, but the *meaning* is profound: we've just proven that despite the randomness in SGD, if we choose our learning rate carefully based on properties of the problem (the singular values), we get guaranteed exponential convergence.

Your approach:

You're right to solve $-c_1\eta + c_2\eta^2 < 0$. Factor out η to get $\eta(-c_1 + c_2\eta) < 0$. Since $\eta > 0$ (we need positive learning rate), this means we need $-c_1 + c_2\eta < 0$, which gives $\eta < c_1/c_2$.

Does this make more sense now about why the whole proof structure was necessary?

Yes, let's go to the next question.

Perfect! Your understanding is correct. The key takeaway from that problem is that for small enough η (specifically $\eta < c_1/c_2$), the beneficial linear term dominates the harmful quadratic term, guaranteeing exponential convergence of SGD.

Here is the complete solution:

Even though I told it to move on, it adhered with its original instructions and outputted the solution first, which was nice.

Solution:

Starting from our inequality:
$E[L(\tilde{w}_{t+1})|\tilde{w}_t] = E[L(\tilde{w}_t)] + E[A|\tilde{w}_t] + E[B|\tilde{w}_t]$

Substituting our bounds from parts (h) and (i):
$E[L(\tilde{w}_{t+1})|\tilde{w}_t] \leq L(\tilde{w}_t) - (4/n)\eta\sigma^2\_minL(\tilde{w}_t) + (4/n)\eta^2\rho^2\sigma^2\_maxL(\tilde{w}_t)$

This simplifies to:
$E[L(\tilde{w}_{t+1})|\tilde{w}_t] \leq (1 - (4/n)\eta\sigma^2\_min + (4/n)\eta^2\rho^2\sigma^2\_max)L(\tilde{w}_t)$

Therefore, we have $1 - c_1\eta + c_2\eta^2 < 1$ if $\eta < c_1/c_2$. To minimize the convergence rate (maximize convergence speed), we can choose $\eta = c_1/(2c_2)$. Note that c_1 and c_2 are positive constants determined by the problem's singular values.

Let's move on to the next question!

Question 3: Problem 3, Part (g)

Context: You're analyzing gradient descent with momentum for the least-squares problem $L(w) = ||y - Xw||^2_2$. The momentum update is:

$w_{t+1} = w_t - \eta z_{t+1}$
$z_{t+1} = (1 - \beta)z_t + \beta g_t$

Through the previous parts, you've discovered something remarkable: by using the SVD of X and reparameterizing into a rotated coordinate system, the problem decouples into independent 2×2 dynamical systems, one for each singular value σ_i. Each 2×2 system has a matrix R_i with eigenvalues that determine convergence.

You've learned that: Again, it did really well summarizing takeaways of previous parts.

- When eigenvalues are repeated or complex conjugates, they both have magnitude $\sqrt{(1-\beta)}$, giving very fast convergence that's independent of the singular values!
- When eigenvalues are distinct and real, one is inside the circle of radius $\sqrt{(1-\beta)}$ and one is outside, giving slower convergence
- The transition happens at specific learning rates determined by the discriminant

Question: Assume that $\beta = 0.1$ and we have a problem with two singular values $\sigma^2\_max = 5$ and $\sigma^2\_min = 0.05$. What learning rate η should we choose to get the fastest convergence for gradient descent with momentum? Compare how many iterations it will take to get within 99.9% of the optimal solution (starting at 0) using this learning rate and momentum with what it would take using ordinary gradient descent.

What is your conceptual approach to solving this problem?

For ordinary gradient descent, we can probably just use the formula for optimal rate and then get the number of iterations using the same method as in 1a. I don't know how to do this for momentum. I only partially knew how to solve this problem, so I asked it for help.

Great! You've got the right idea for ordinary gradient descent. Let me help you build up the intuition for the momentum case, because it connects beautifully to what you learned in the earlier parts.

For Ordinary Gradient Descent:

You're exactly right. From part (e) of Problem 1, you know the optimal learning rate is $\eta^* = 1/(\sigma^2\_min + \sigma^2\_max)$, and the optimal convergence rate is $(\sigma\_max/\sigma\_min)^2 - 1$ divided by $(\sigma\_max/\sigma\_min)^2 + 1$. With $\sigma^2\_max = 5$ and $\sigma^2\_min = 0.05$, you have $\sigma\_max/\sigma\_min = 10$, so the ratio of their squares is 100. This gives you a convergence rate, and then you can find how many iterations T you need by solving (rate)^T ≤ 0.001 (to get within 99.9% means the error is at most 0.1% of the optimum).

For Gradient Descent with Momentum:

Here's where it gets interesting, and this is why you did all that eigenvalue analysis in parts (d) through (f). Remember the key insight: you want the eigenvalues of each 2×2 system matrix R_i to either be repeated real or complex conjugate pairs, because in those cases they have magnitude exactly $\sqrt{(1-\beta)} = \sqrt{0.9} \approx 0.948$. This is your convergence rate, and it's *independent of the singular values*!

The question is: can you find a single learning rate η that puts *both* singular values ($\sigma^2\_max = 5$ and $\sigma^2\_min = 0.05$) into this "sweet spot" where their eigenvalues are either repeated or complex?

From part (f), you found that eigenvalues are complex (or repeated) when the learning rate satisfies:

$(2 - \beta - 2\sqrt{(1-\beta)})/(2\beta\sigma^2_i) \leq \eta \leq (2 - \beta + 2\sqrt{(1-\beta)})/(2\beta\sigma^2_i)$

The crucial question: is there an overlap in the valid η ranges for both σ^2_max and σ^2_min? If yes, any η in that overlap gives you the magical $\sqrt{(1-\beta)}$ convergence rate for both dimensions simultaneously!

I get the feeling that the hint it gave was too strong. The actual solution has more steps.

Your approach:

Calculate the valid η interval for $\sigma^2\_max = 5$, calculate the valid η interval for $\sigma^2\_min = 0.05$, and check if they overlap. If they do, pick any η in the overlap (the solution picks one in the middle of the range), and your convergence rate is simply $\sqrt{0.9}$, regardless of which dimension you're looking at. Then compare the number of iterations needed.

Can you see why this is so powerful? With momentum, you've essentially "equalized" the convergence rates across dimensions!

Ok, then I guess we can just plug in the values we have and pick the tightest bound. When η is in that bound, we can get the convergence rate and then use that to compute the number of iterations similarly.

Perfect! You've got it exactly right. Your conceptual approach is spot on.

Let me confirm your understanding: you'll compute the valid η interval for each singular value using the formula from part (f), find where they overlap (the intersection of the two intervals), and any η in that overlap guarantees that both dimensions have eigenvalues with magnitude √(1-β) = √0.9 ≈ 0.948. Then you use this convergence rate to find the number of iterations needed to reduce the error by a factor of 1000 (to get within 99.9%, or 0.1% error).

The beautiful insight here is that when you compare this with ordinary gradient descent, you'll see that momentum can be dramatically more efficient. With ordinary gradient descent, the large condition number (the ratio σ_max/σ_min = 10, or squared ratio of 100) really hurts you because one dimension wants a small learning rate while the other can handle a large one. You're stuck with a compromise that makes the convergence rate close to 1 (very slow). But with momentum, you've essentially decoupled this tradeoff - both dimensions can converge at the same favorable rate √0.9, which is noticeably smaller than what ordinary gradient descent achieves.

Again, I really like how it elaborates on implications, even with no prompt to do this.

Here is the complete solution:

Solution:

For ordinary gradient descent, using equation (85), the optimal convergence rate is $R_1 = (5-0.05)/(5+0.05) = 0.98$. Therefore, the minimum number of iterations T_1 to get within 99.9% of the optimal solution is:

$||w\_T_1 - w^*||^2 = R_1{}^\wedge T_1\ ||w_0 - w^*||^2 \leq 0.001||w^*||^2$

Solving: $T_1 = \lfloor \log(1000)/\log(505/495) \rfloor = 346$ iterations

For gradient descent with momentum, we need to find the valid η ranges where eigenvalues are complex or repeated. Using β = 0.1, we calculate:

For $\sigma^2\_max = 5$: $0.002633 \leq \eta \leq 3.797$
For $\sigma^2\_min = 0.05$: $0.2633 \leq \eta \leq 379.7$

The intersection (common interval where both dimensions have complex/repeated eigenvalues) is: $0.2633 \leq \eta \leq 3.797$

Any η in this range gives convergence rate $R_2 = \sqrt{(1-\beta)} = \sqrt{0.9} = 0.949$

With this convergence rate: $T_2 = \lfloor \log(1000)/\log(1/0.949) \rfloor = 132$ iterations

Comparison: Momentum requires only 132 iterations compared to 346 for ordinary gradient descent - approximately 2.6× faster! This demonstrates how momentum can dramatically accelerate convergence by allowing both dimensions to converge at the favorable rate $\sqrt{(1-\beta)}$, effectively neutralizing the impact of poor conditioning.

Ready for the next question?

Yes. Again, wasting a message.

Question 4: Problem 5, Part (a)

Context: You're analyzing data augmentation through instance noise. You have m labeled data points (x_i, y_i) where each $x_i \in \mathbb{R}^n$ and each $y_i \in \mathbb{R}$. During training with SGD, you add random Gaussian noise to each input vector every time you use it. This means the effective input seen by SGD is:

$\check{X}_i = x_i + N_i$

where the noise vectors N_i are distributed as $N_i \sim N(0, \sigma^2 I_n)$ and are independent and identically distributed.

You arrange these noise-augmented data points into a random matrix $\check{X} \in \mathbb{R}^{m \times n}$ with rows $\check{X}_i^T$, and the labels into a vector y.

Question: Show that the optimization problem

argminw $E[||\check{X}w - y||^2]$

is equivalent to a regularized least squares problem:

argminw $(1/m)||Xw - y||^2 + \lambda||w||^2$

You need to determine the value of λ. The hint suggests: write the squared norm of a vector as an inner product, expand, and apply linearity of expectation.

What is your conceptual approach to showing this equivalence?

If you expand out the expectation, you can separate it into the original least squares expression and something that resembles regularization, given that the random noise is known to be $\sigma^2 I_n$.

Excellent! You've identified the key insight perfectly. Your approach is exactly right - when you expand the expectation and use the properties of the Gaussian noise, the cross-terms involving the noise will contribute an additional term that looks exactly like ridge regularization.

Let me walk through why this works conceptually before you see the full solution. The magic happens because of three properties of the noise:

First, when you expand $||\check{X}w - y||^2$, you'll get three types of terms: terms involving only the true data X (which give you the original least squares objective), cross-terms mixing the noise N with the data, and terms involving only the noise N.

Second, the noise N_i has zero mean, so $E[N_i] = 0$. This means when you take expectations, all those cross-terms like $E[N_i^T w(x_i^T w - y_i)]$ vanish because the noise is independent of everything else and has zero expected value.

Third, even though the noise has zero mean, it has variance σ^2! So when you look at the quadratic noise term $E[(N_i^T w)^2]$, this doesn't vanish. Instead, using the fact that $E[N_i N_i^T] = \sigma^2 I_n$ (the covariance matrix of the noise), you get $E[(N_i^T w)^2] = E[w^T N_i N_i^T w] = w^T E[N_i N_i^T]w = \sigma^2 w^T w = \sigma^2||w||^2$.

When you sum this over all m data points, you get exactly $m\sigma^2||w||^2$ added to your original least squares objective. Dividing through by m to match the form in the problem gives you $\lambda = \sigma^2$.

This is a profound result! It tells us that adding random noise to inputs during training is mathematically equivalent to explicit L2 regularization. This explains why data augmentation helps prevent overfitting - it's secretly doing regularization. The amount of regularization (λ) is directly controlled by how much noise variance (σ^2) you add.

Your conceptual understanding is spot on. Here's the complete solution:

Solution:

Following the hint, we write our objective as:

$E[||\check{X}w - y||^2] = E[\sum_{i=1}^m (\check{x}_i^T w - y_i)^2]$

By linearity of expectation:
$= \sum_{i=1}^m E[((x_i + N_i)^T w - y_i)^2]$

Expanding the square:
$= \sum_{i=1}^m E[(x_i^T w - y_i)^2 + 2(N_i^T w)(x_i^T w - y_i) + (N_i^T w)^2]$

Applying linearity of expectation and using $E[N_i] = 0$ (which makes the cross-term vanish):
$= \sum_{i=1}^m [(x_i^T w - y_i)^2 + 0 + E[w^T N_i N_i^T w]]$

Using $E[N_i N_i^T] = \sigma^2 I_n$:
$= \sum_{i=1}^m [(x_i^T w - y_i)^2 + w^T \sigma^2 I_n w]$

$= ||Xw - y||^2 + m\sigma^2||w||^2$

Dividing through by m to match the desired form gives us:

argminw $(1/m)||Xw - y||^2 + \sigma^2||w||^2$

Therefore, **$\lambda = \sigma^2$.**

This remarkable result shows that instance noise augmentation is mathematically equivalent to L2 regularization, with the regularization strength directly proportional to the noise variance.

Ready for the final question?

Yes

No comments about this, I've done this problem before.

Question 5: Problem 7

Context: You're exploring an alternate probabilistic interpretation of Ridge Regression. You know the standard Ridge Regression solution is:

$\hat{w} = (X^T X + \lambda I)^{-1} X^T y$ (the "primal form")

And there's an alternate form called the Kernel Ridge form:

$\hat{w} = X^T (X X^T + \lambda I)^{-1} y$ (the "dual form")

You've already seen that Ridge Regression can be interpreted as finding the Maximum A Posteriori (MAP) estimate when you put a Gaussian prior $W \sim N(0, I)$ on the weights and model $Y = XW + \sqrt{\lambda}N$ where $N \sim N(0, I)$.

Now, the problem sets up a joint probability perspective. The underlying probability space includes d independent standard normals that define W and n independent standard normals that define the noise on the n training points. The data matrix X is fixed (not random). This means W and Y are jointly normal random vectors.

Given facts:

- The primal form $(X^T X + \lambda I)^{-1} X^T y$ is the MAP estimate for W given observation $Y = y$
- For jointly normal random variables, conditioning preserves normality
- A normal random variable has its density maximized at its mean
- For jointly normal zero-mean random vectors, the conditional expectation formula is: $E[W|Y = y] = \Sigma_{WY} \Sigma_{YY}^{-1} y$, where $\Sigma_{YY} = E[YY^T]$ is the covariance of Y and $\Sigma_{WY} = E[WY^T]$ is the cross-covariance

Question: Use these facts to show that the two forms of the Ridge Regression solution are identical by computing the conditional expectation $E[W|Y = y]$ and showing it equals the Kernel Ridge form.

What is your conceptual approach to proving these two solutions are the same?

To find E[W|Y=y], we can solve the covariances independently by plugging in Y, expanding, and computing intermediate values. This probably results in an expression that looks the same as the objective for ridge regression.

I gave a super vague answer and it still proceeded. I think I would've liked if it pushed for a more specific answer, but this is on me for not providing a specific enough prompt.

Perfect! You've got the right conceptual approach. You're going to compute the two covariance matrices (Σ_{WY} and Σ_{YY}) separately by expanding them out, using the structure of the generative model $Y = XW + \sqrt{\lambda}N$, and then plug them into the conditional expectation formula.

The key insight you're touching on is that when you compute these covariances, the algebra will naturally produce the Kernel Ridge form $X^T(XX^T + \lambda I)^{-1}y$. This will show that the MAP estimate (which you know from earlier work is the primal form) must equal the conditional expectation (which you'll show is the dual form), proving they're identical.

Let me clarify one small point about your answer: the expression you get won't look like the "objective" for ridge regression (that's the loss function we minimize), but rather it will look exactly like the dual solution formula - the weights themselves.

Here's the beautiful logic of the proof: Since W and Y are jointly Gaussian and zero-mean, when you condition W on Y = y, you get another Gaussian distribution. For Gaussian random variables, the MAP estimate (mode) equals the mean of the distribution. So $MAP(W|Y = y) = E[W|Y = y]$. You already know MAP equals the primal form from previous work, so now you just need to compute $E[W|Y = y]$ using the covariance formula and show it gives the dual form.

The computation itself follows your approach exactly:

For Σ_{WY}: You'll expand $E[WY^T] = E[W(XW + \sqrt{\lambda}N)^T]$, distribute, and use the facts that $E[WW^T] = I$ (from the prior), $E[W] = 0$, $E[N] = 0$, and that W and N are independent. This will give you $\Sigma_{WY} = X^T$.

For Σ_{YY}: You'll expand $E[YY^T] = E[(XW + \sqrt{\lambda}N)(XW + \sqrt{\lambda}N)^T]$, distribute to get four terms, and again use independence and the covariance structures. The cross-terms vanish (zero mean), and you're left with $XE[WW^T]X^T + \lambda E[NN^T] = XX^T + \lambda I$.

Finally: Plugging these into the formula $E[W|Y = y] = \Sigma_{WY} \Sigma_{YY}^{-1} y$ gives you $X^T(XX^T + \lambda I)^{-1}y$, which is exactly the Kernel Ridge form!

Your conceptual understanding is excellent. Here's the complete solution:

Solution:

We are given that the primal form $(X^TX + \lambda I)^{-1}X^Ty$ equals $MAP(W|Y = y)$. Let's find $MAP(W|Y = y)$ using a different approach.

When we condition the random variables W on the values Y = y, the resulting conditional distribution $W|Y = y$ is still normal (since they're jointly normal). Since a normal random variable has its density maximized at its mean, the MAP estimate equals the conditional expectation: $MAP(W|Y = y) = E[W|Y = y]$.

We'll use the formula $E[W|Y = y] = \Sigma_{WY} \Sigma_{YY}^{-1} y$. First, let's calculate $\Sigma_{WY} = E[WY^T]$.

Using $Y = XW + \sqrt{\lambda}N$:
$\Sigma_{WY} = E[W(XW + \sqrt{\lambda}N)^T] = E[WW^TX^T] + \sqrt{\lambda}E[WN^T]$

Since W and N are independent and both zero-mean, $E[WN^T] = E[W]E[N^T] = 0$. Also, $E[WW^T] = I$ (the covariance of W). Therefore:

$\Sigma\_WY = IX^T = X^T$

Now let's find $\Sigma\_YY = E[YY^T]$:
$\Sigma\_YY = E[(XW + \sqrt{\lambda}N)(XW + \sqrt{\lambda}N)^T]$
$= E[XWW^TX^T] + \lambda E[NN^T] + \sqrt{\lambda}E[XWN^T] + \sqrt{\lambda}E[NW^TX^T]$

The cross-terms vanish since W and N are independent with zero means. Using $E[WW^T] = I$ and $E[NN^T] = I$:
$\Sigma\_YY = XE[WW^T]X^T + \lambda I = XX^T + \lambda I$

Finally, plugging into the conditional expectation formula:
$\hat{w} = E[W|Y = y] = \Sigma\_WY\, \Sigma\_YY^{-1}\, y = X^T(XX^T + \lambda I)^{-1}y$

This is exactly the Kernel Ridge form! Since MAP(W|Y = y) equals both the primal form and E[W|Y = y] (which we just showed equals the dual form), the two solutions are identical.

This elegant result shows that the primal and dual forms of Ridge Regression are not just algebraically equivalent - they represent the same probabilistic quantity (the conditional mean/MAP estimate) viewed from different perspectives.

Excellent work! You've now reviewed the 5 most important conceptual problems from this homework. These covered: optimal learning rates and condition numbers, stochastic convergence theory with Lyapunov functions, momentum-based acceleration, the equivalence of data augmentation and regularization, and probabilistic interpretations of ridge regression. These are foundational concepts that will serve you well throughout your deep learning studies. Good luck on your exam!

Really good summary of what I learned, and I like that Claude is always super encouraging.